



Field measurements reveal local limits of imputed functional traits in tropical trees

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Abstract

This study evaluated the reliability of functional data imputed in the classification of ecological strategies according to the CSR model (competitive, stress-tolerant, ruderal) in tree species of a riparian Atlantic Forest in Brazil. Empirical data collected in situ from 68 species were compared with data imputed using the Bayesian Hierarchical Probabilistic Matrix Factorization (BHPMF) technique, considering key functional attributes such as leaf area and specific leaf area. The results showed significant divergences between the classifications based on measured and imputed data, especially for the traits directly involved in the formulation of the CSR model. While imputation overestimated selection for stress tolerance, field data indicated a greater diversity of strategies, with a greater presence of competitive species. The discrepancies point to limitations in the exclusive use of imputed data in local assessments of functional diversity, highlighting the need to integrate empirical collections and improvements in imputation methods for applications in environmental management and restoration.

Keywords Atlantic forest · Data imputation · Missing data · Plant strategy · Scale-dependence

Introduction

Plant functional traits are a subset of plant characteristics that correspond specifically to measurable attributes related to ecological functions. These traits (which can be morphological, anatomical, biochemical, physiological, and/or phenological) are quantified at the individual level and reflect how plants interact with the environment and respond to ecological and evolutionary pressures (Wright et al. 2004). Because functional traits are explicitly linked to plant performance and ecosystem processes, they are widely used to compare ecological strategies among species and communities (Violle et al. 2007).

Integrating these traits, and how they trade-off against one another, is crucial for understanding how plants are adapted to environmental pressures and interact with their ecosystem, offering a solid foundation for exploring species richness and functional diversity within ecosystems (Thomas et al. 2020). As the relative investment of resources in different functions has absolute limits that are common across the world, quantifying functional characteristics allows ecological comparisons between floras from distinct ecosystems and regions (Díaz et al. 2016; Pierce et al. 2017).

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One system for classifying functional groups and assigning a quantitative position along a trade-off spectrum is the scheme of CSR (competitor, stress-tolerator, ruderal) ecological strategies (Grime 1977). According to the CSR scheme, sets of functional traits consistently evolve in response to different degrees of competition for resources (resource pre-emption), abiotic stress (limitations to metabolic efficiency), and frequent disturbances that lead to biomass destruction (ruderal). These sets of traits can consist of characters that are easily measured and correlated with the main functional abilities of plants, which allows the replication and comparison of results in different communities (e.g., Negreiros et al. 2014; Pierce et al. 2017; Rosenfield et al. 2019).

Among the traits most closely linked to vegetation modeling parameters, CSR theory (Grime and Pierce 2012) outlines a functional trade-off triangle defined by variation in specific leaf area (SLA), leaf dry matter content (LDMC), and leaf area (LA). These three attributes represent key axes of ecological strategies across global flora, delineating a spectrum from resource-acquisitive to resource-conservative species. Target species occupy specific positions within this trade-off triangle, reflecting evolutionary compromises between maximizing light interception (via LA and SLA), growth potential (SLA), and tissue longevity or structural investment (LDMC). These traits are functionally associated with vegetation carbon storage (height), light capture efficiency (SLA and LA), and nutrient cycling, which is driven by photosynthetic capacity and the extent of light interception enabled by leaf area and mass (see Aerts and Chapin 1999; Lavorel and Garnier 2002; Díaz et al. 2004; Garnier and Navas 2012).

Based on standardized protocols for measuring these functional traits in situ (Cornelissen et al. 2003; Pérez-Harguindeguy et al. 2013), databases gathering all this information are available to the entire scientific community (Fitter and Peat 1994; Poschlod et al. 2003; Baraloto et al. 2010). Among the available databases, TRY (version 6.0) stands out as the most comprehensive. However, despite being the largest database we have so far, it still has a significant percentage of missing data, even for well-documented attributes such as specific leaf area, vegetative height, leaf dry matter content, and seed mass (Kattge et al. 2011, 2020). This high number of missing data limits statistical analyses of functional diversity (e.g., CSR) to only the most abundant species (Taugourdeau et al. 2014).

One technique that has emerged as a promising alternative to overcome the limitations of in situ data collection is gap-filling (or imputation) (Penone et al. 2014). Imputation makes it possible to estimate missing or incomplete values of functional traits based on statistical models, such as Bayesian hierarchical probabilistic matrix factorization

(BHPMF) (Schrodte et al. 2015). Currently, BHPMF is one of the most robust gap-filling tools that consider the phylogenetic signal of the trait and the correlation between traits (Joswig et al. 2023). In addition to BHPMF, recent studies have applied different metrics to assess the accuracy and ecological implications of trait imputation. Stewart et al. (2023) tested the robustness of functional diversity metrics under increasing levels of missing data, applying different imputation strategies and showing that richness (FRic), divergence (FDiv), and weighted functional dissimilarity (Rao's Q) can remain representative even with 50–70% missing data. Complementarily, Rodríguez-Alarcón et al. (2022) explicitly evaluated how incorporating intraspecific trait variation (ITV) influences functional diversity metrics, including functional richness (FRic) and trait probability density (TPD), demonstrating that neglecting ITV can obscure ecologically relevant functional patterns at local scales. Liu et al. (2025), in turn, combined trait imputation with global-scale biodiversity models, assessing differences in functional richness and conservation priorities derived from measured versus imputed data, and warning that imputations can inflate diversity estimates and bias large-scale policy decisions. These approaches complement trait-level validation methods, such as BHPMF (Schrodte et al. 2015), which typically rely on cross-validation using root mean square error (RMSE) and coefficient of determination (R^2) to assess predictive accuracy, but do not explicitly account for ecological consequences at community or biogeographic scales.

Although gap-filling is not intended to replace in situ data collection (Guo et al. 2022; Joswig et al. 2023), the indiscriminate use of estimated or imputed values in functional diversity analyses can distort the real information in the data and consequently result in erroneous conclusions. In this study, we test the hypothesis that ecological strategies, as defined by CSR theory, remain similar when derived from imputed versus directly measured characteristic data.

Materials and methods

Empirical data

To respond to our question, we selected 68 tree species from the phytosociological table presented in Ramos et al. (2023) (Table S1). The study was conducted in remnants of riparian Atlantic Forest along 473 km of the Rio Doce watershed, covering five municipalities between Mariana and Aimorés, Minas Gerais, Brazil. The climate of the forest remnants studied are humid subtropical with dry winters and hot summers (Cwa) and seasonal tropical with dry winters and rainy summers (Aw) (Köppen's classification) (Alvares et

al. 2013). The sampling sites were carefully chosen in relatively well-preserved riparian forest areas. Details on the sampled sites, their environmental characteristics, and land use patterns are described in Ramos et al. (2023).

Functional trait data were obtained from the local dataset reported by Justino et al. (2025), following Pérez-Harguindeguy et al. (2013). Branches were collected from five individuals of each of the 68 tree species. In the field, branches were identified, moistened, and stored in plastic bags until reaching the laboratory. To prevent cavitation and rehydrate the leaves to turgidity, branches were placed in buckets of water with 5 cm of the stem cut off. Both the buckets and branches were kept in the dark for a minimum of 2 h (see Pérez-Harguindeguy et al. 2013). After hydration, three mature leaves with no sign of herbivory or fungus infection from each branch were selected to measure leaf attributes. All leaves analyzed included the petioles and in the case of compound leaves, the rachises and rachillae were considered in the analysis. With the leaf completely turgid and using a precision analytical scale (0.1 mg), the leaf fresh weight was measured, after which the leaf was scanned using a desktop scanner and processed in the ImageJ software to measure the leaf area. To obtain dry weight, all leaf material was dried in an oven at 70 °C for at least 72 h and then weighed (see Pérez-Harguindeguy et al. 2013). The methodological approach adopted here follows standardized and widely recognized protocols for functional trait measurement (Cornelissen et al. 2003; Pérez-Harguindeguy et al. 2013), ensuring consistency with most trait-based ecological studies. This alignment allows our in situ measurements to be directly comparable with data compiled in global trait databases, including those used by Guo et al. (2022). Leaf area (LA) corresponded to projected turgid leaf area, specific leaf area (SLA) was calculated as leaf area divided by dry mass, and leaf dry matter content (LDMC) as dry mass divided by turgid fresh mass. For each species, trait values were calculated from leaf-level measurements and then averaged across three leaves per individual and five individuals per species to obtain species-level values. To categorize the species into functional types according to the CSR theory, we used the “StrateFy” spreadsheet, a globally calibrated analysis tool that calculates CSR coordinates and classifies the ecological strategy of each species based on the average values of SLA, LDMC, and LA (Pierce et al. 2017). This method does not assign C-, S-, and R-selection scores from single traits independently, but calculates them from the trade-off among LA, SLA, and LDMC, using regression equations calibrated against the global trait trade-off (Pierce et al. 2017). Thus, the resulting CSR scores summarize how each species’ combination of leaf size, leaf economics, and tissue investment compares with global patterns of plant ecological strategies.

Although wood density (WD) was not directly used in the analysis of CSR ecological strategies, this functional trait was also available in the empirical dataset reported by Justino et al. (2025). For WD measurement, the cut stems were peeled with a stylus, separated by species and placed in smaller plastic jars to hydrate for at least 1 h. The saturated volume of the stem was measured using a jar of water, a needle and a scale, following the water displacement method, in which the mass of displaced water (g) corresponded to the saturated volume (cm³). Subsequently, stem segments were dried in an oven at 70 °C until constant mass and weighed to obtain dry mass. Wood density (g cm⁻³) was calculated for each individual as dry mass divided by saturated volume, and the species-level WD value corresponded to the arithmetic mean of the five sampled individuals (see Pérez-Harguindeguy et al. 2013).

Gap filling and data analysis

To meet the objective of evaluating the reliability of imputed functional trait data in ecological strategy classification, we selected the 68 species from the phytosociological table presented in Ramos et al. (2023) that were also listed in Guo et al. (2022) and for which imputed trait values were available. We did not perform a new imputation procedure; instead, the imputed trait values used in this analysis were extracted directly from the supplementary material published by Guo et al. (2022). To calculate CSR classes from the imputed dataset, the LA, SLA, and LDMC values extracted for the 68 species were entered into the “StrateFy” spreadsheet to obtain proportional C, S, and R scores and the corresponding ecological strategy classification, following the same procedure applied to the empirical dataset (Pierce et al. 2017). Guo et al. (2022) compiled 21 functional traits from the TRY, TOPIC, and BIEN databases, and due to the absence of some values, they applied the Bayesian hierarchical probabilistic matrix factorization technique (BHPMF - a machine learning technique that considers phylogenetic trait signal and trait-trait correlations; Schrodte et al. 2015). This method factors the species-specific matrix into lower-dimensional latent matrices using Bayesian priors that integrate phylogenetic feature signals and feature-feature correlations, thereby improving the accuracy of predictions. In practice, BHPMF ensures that phylogenetically related species and ecologically correlated characteristics constrain the estimation of missing values, outperforming simpler imputations based only on phylogeny or mean values. The proportion of missing values filled by BHPMF was high, since 98.53% of species had imputed values for LDMC, 88.13% for WD, 98.52% for LA, and 96.36% for SLA (Guo et al. 2022). To evaluate the imputed dataset, the authors used the beanplot package (Kampstra 2015) to compare imputed and observed

data, as well as the `phylosig` function in the `phytools` package (Revell 2012) to identify which of the characteristics presented significant phylogenetic signals.

To assess the congruence between the values of the characteristics measured from the material collected in situ and the values derived from the imputation technique, simple linear correlation was used, considering a significance level (α) of 5% (Quinn and Keough 2002). Data distribution was assessed, and logarithmic transformation was applied to both imputed and measured LA to satisfy normality assumptions.

Results

Based on the data collected in situ, the imputation method resulted in overestimation of the extent of S-selection and underestimation of both C- and R-selection (Fig. 1). Of the 68 species, only *Aniba firmula* (Lauraceae), *Magonia pubescens* (Sapindaceae), *Miconia latecrenata* (Melastomataceae), *Siparuna guianensis* (Siparunaceae) and *Tabernaemontana solanifolia* (Apocynaceae) showed identical functional categories between the two methods. Conversely, when considering the relative percentages of C: S:R, we found significant variation between the two methods (Table S1).

For several species, the discrepancy between CSR categorizations based on imputed and measured data was substantial, with 18 species exhibiting entirely divergent ecological strategies (Fig. 1, Table S1). For example, *Dalbergia nigra* (Fabaceae) showed extreme differences, being categorized as stress-tolerant (S: 87.3%) based on imputed values, whereas field data indicated a balanced competitive-ruderal profile (C: 37.5%, R: 38.0%). A notable shift also occurred in *Chrysophyllum gonocarpum* (Sapotaceae), which transitioned from stress-tolerant in situ (S: 63.6%) to predominantly competitive in the imputed classification (C: 57.6%). These cases underscore how imputation can significantly distort ecological interpretations when not grounded in species-specific empirical data.

Contrary to our hypothesis, the comparison between imputed and in situ data revealed that not all functional traits used to define ecological strategies are equally reliable when estimated. Significant correlations were observed for LDMC and WD, indicating a reasonable agreement between imputed and measured values for these traits. In contrast, key traits directly involved in CSR classification, LA and SLA showed no correlation between methods. Likewise, there was no agreement in the derived C, S, and R scores, highlighting that imputed data may fail to reflect key functional dimensions underlying ecological strategies (Fig. 2, Table S1).

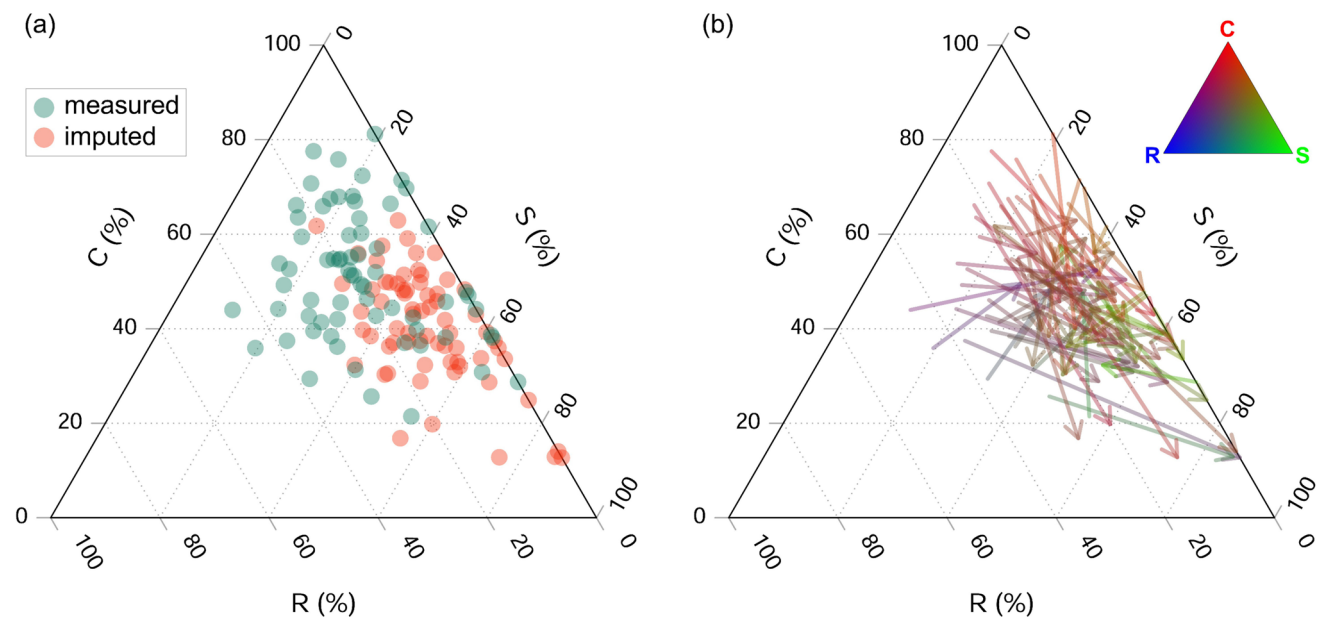
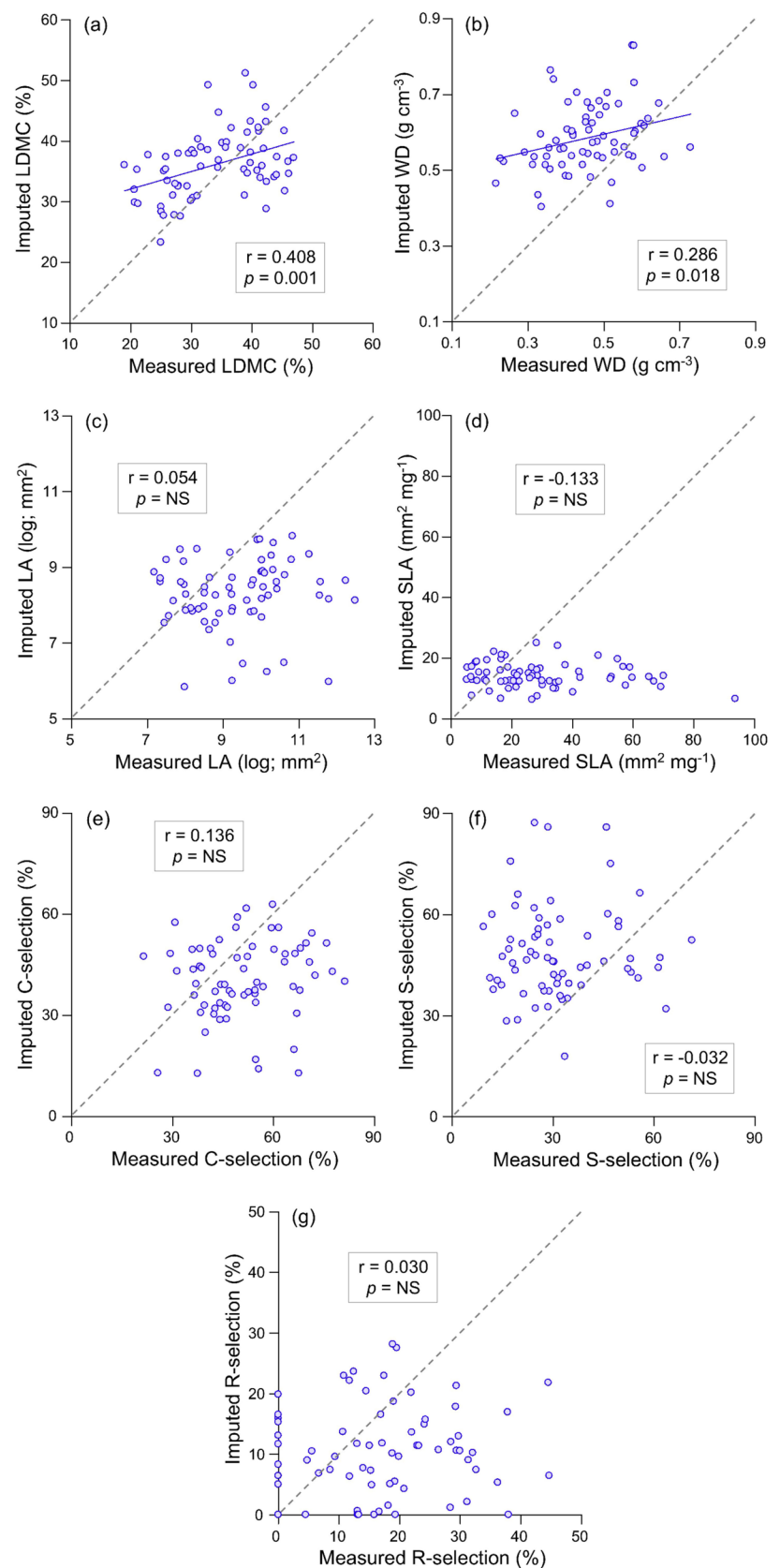


Fig. 1 Relative importance of plant CSR strategies showing ternary plots with the degree of C-selection (competitiveness), S-selection (stress-tolerance), and R-selection (ruderality) for 68 tree species. **(a)** Green circles represent CSR derived from empirically measured traits from species occurring in tropical riverine ecosystem in southeastern Brazil, and orange circles represent CSR derived from data imputed with gap filling methods, obtained from Guo et al. (2022). **(b)** Arrows indicating the shift of C-, S- and R- selection from measured data

(arrow origin) to imputed data (arrow tip). The color of each arrow corresponds to the conversion of the ratio of empirically measured C, S, and R to red, green, and blue, respectively (i.e., the color of arrow origin). Top right panel shows the color guide for the arrows. The most dramatic changes in strategies were shown by *Senegalia langsdorffii* (54.7% of change in C-selection degree), *Dalbergia nigra* (62.7% of change in S-selection), and *Pogonophora schomburgkiana* (38.2% of change in R-selection)

Fig. 2 Relationship between empirically measured traits and data imputed by gap filling methods for 68 tree species. LDMC: leaf dry matter content; WD: wood density; LA: leaf area; SLA: specific leaf area; C-selection: degree of competitive strategy; S-selection: degree of stress-tolerance strategy; R-selection: degree of ruderality. For each panel, it is shown the correlation coefficient (r) and the significance (p). Dashed gray line indicates perfect correspondence (i.e., $r=1$) between measured and imputed trait values. Best fit line is shown only for significant relationships



Discussion

The categorization of species into ecological strategies of CSR showed that the empirical data indicated a balance between conservative and acquisitive resource-use strategies, with a slight tendency toward the competitive vertex of the CSR triangle. In contrast, imputed data shifted species toward higher S-selection and lower C- and R-selection, contradicting our hypothesis that CSR strategies remain consistent regardless of the data collection method. This divergent distribution of species along the Grime triangle (Pierce et al. 2013, 2017) highlights a central dilemma in contemporary functional ecology, which consists of balancing the scope of large-scale analyses with the accuracy of ecological interpretations at the local level.

Macroscale studies, such as that by Guo et al. (2022), which maps the exposure of tree diversity to anthropogenic pressures, are intrinsically dependent on filling gaps to be viable. The robustness of this approach is supported by research such as that of Stewart et al. (2023), who demonstrated that functional diversity metrics can remain representative even with up to 50–70% of data missing, if imputation is used judiciously. This ability to generate complete datasets allows for significant advances in understanding how environmental factors shape biodiversity on a planetary scale.

However, although these methods increase data coverage and allow for global analyses, they can obscure ecologically relevant variation at the local level (Penone et al. 2014; Sandel et al. 2015). Our comparative analysis revealed significant divergences between ecological strategies inferred from imputed data and those derived from field measurements, especially for traits such as SLA and LA, showing no agreement between the methods. This low congruence is likely influenced by the very high proportion of missing data in the source database for these traits, since lower levels of missingness would be expected to improve the agreement between imputed and empirically measured values. In addition, imputed species-level values may mask intraspecific variation in traits (ITV), a crucial component of local adaptation and community dynamics. As shown by Rodríguez-Alarcón et al. (2022), relying solely on species-level averages becomes problematic when individual variability is high, particularly for traits linked to environmental heterogeneity. It is also important to note that the trait database compiled by Guo et al. (2022), from which imputed values were extracted for comparison, does not provide information on data completeness at the species level, preventing direct identification of whether individual trait values are empirical or imputed. Nonetheless, the very high proportions of missing data reported for the traits underlying CSR classification (96–99% for SLA, LDMC, and LA) indicate

that imputation necessarily represents a dominant component of these trait estimates. While part of the observed divergence may reflect genuine phenotypic variation across regions and environments, these results highlight the limitations of treating imputed values as functional equivalents of empirical measurements when interpreting plant strategies at local scales.

In addition, wood density (WD) revealed a distinct pattern compared to leaf traits. While SLA and LA showed marked discrepancies between empirical and imputed data, WD was more consistent between methods. This pattern is consistent with previous study recognizing WD as a structurally conservative trait, strongly integrated into the wood economics spectrum (Chave et al. 2009), rather than necessarily indicating better imputation performance. This interpretation is also consistent with evidence from tropical dry forests showing that incorporating individual-level trait variation, including stem-related traits, does not substantially improve trait-growth relationships relative to models based on species-level averages (Rodríguez-Alarcón et al. 2024). Consequently, even when estimated using BHPMF, WD maintained functional patterns close to those derived from field observations, confirming that characteristics with greater evolutionary conservatism are more suitable for macroecological studies based on gap-filling approaches (Guo et al. 2022). However, our results emphasize that direct measurements remain essential for capturing local and context-dependent variations, particularly in heterogeneous tropical environments, where failure to capture subtle intraspecific differences can lead to significant misinterpretations of a species' ecological strategy.

Discrepancies between CSR classifications based on empirical and imputed data may substantially affect ecological interpretation at the local scale. An emblematic example is *Dalbergia nigra*, classified using imputed data as a purely stress-tolerant species (S: 87.3%), while empirical data indicated a balanced competitive-ruderal profile (C: 37.5%; R: 38.0%). A similar discrepancy was observed for *Chrysophyllum gonocarpum*, which was classified as stress tolerant based on empirical data (S: 63.6%), but as predominantly competitive when imputed data were used (C: 57.6%). These examples do not indicate that one CSR category is a fixed property of the species across its entire distribution. Rather, they show that, within the same local assemblage and using the same CSR classification procedure, empirical and imputed trait values can lead to markedly different functional interpretations. In applied contexts, such shifts could affect how species are interpreted in relation to ecological succession, disturbance responses, and potential performance under local environmental conditions.

When extrapolated to practice, these inaccuracies may compromise the success of conservation and restoration

initiatives. The literature warns of the risks of extrapolating large-scale inferences without considering local ecological heterogeneity (Cadotte et al. 2011; Albert et al. 2012; Violle et al. 2012; Liu et al. 2025). This limitation becomes particularly relevant when results derived from global assessments are used to inform local management and conservation decisions. Overestimating functional diversity, for example, can lead to misallocation of resources, prioritizing areas that are data artifacts, or generating a false sense of security that conservation goals have already been achieved, reducing the political urgency for creating new protected areas (Liu et al. 2025). In our study, this applied implication is illustrated by the displacement of species within the CSR triangle when imputed rather than measured data were used. Imputed values shifted species toward higher S-selection and lower C- and R-selection, meaning that the same species could be interpreted as more stress-tolerant, less competitive, or less ruderal depending on the data source. Such shifts can affect how species' functional strategies are interpreted in local restoration and management contexts.

Given this scenario and considering that climate change and anthropogenic pressures are reshaping ecosystems, it becomes even more crucial to integrate contextually accurate functional data into large-scale projections. In line with recent evidence (Stewart et al. 2023; Liu et al. 2025), our results reinforce the need to apply imputation critically and sensitively to scale, incorporating intraspecific variability and exploring regional adjustment factors to reduce biases. Only then will it be possible to reconcile macroecological analyses with local management and conservation decisions (Cadotte et al. 2011; Kattge et al. 2020; Guo et al. 2022).

Conclusion

Our study aimed to evaluate the reliability of imputed functional data in the classification of ecological strategies according to the CSR model. The results demonstrate that, although the imputation technique expands the coverage of functional data and enables large-scale analyses, it does not accurately replace empirical data in local contexts. We observed marked divergences between imputed values and those measured in the field for essential attributes, such as leaf area and specific leaf area, which resulted in distinct classifications of ecological strategies for a significant number of species. These results indicate that the objective of the study was achieved by highlighting the limits of the exclusive use of imputed data in local functional assessment. As a future perspective, we reinforce the importance of strengthening the integration between field collection efforts and modeling approaches, in addition to encouraging the improvement of imputation methods, especially with

regard to local ecological sensitivity. Enriching databases with empirical information remains essential to ensure more robust ecological inferences that are applicable to environmental management and restoration contexts.

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Author contributions WSJ, DN and GWF wrote the main manuscript text, DN made the statistical analyses and prepared the figures. WSJ, LRA and IZS collected field data. All authors reviewed the manuscript.

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Data availability Data is provided within the manuscript or supplementary information files.

Declarations

Competing interests The authors declare no competing interests.

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